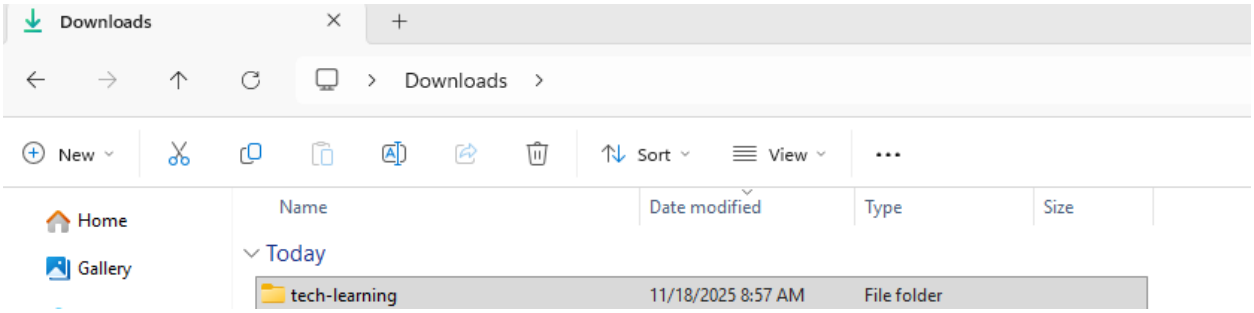
Task 1.1: First Report

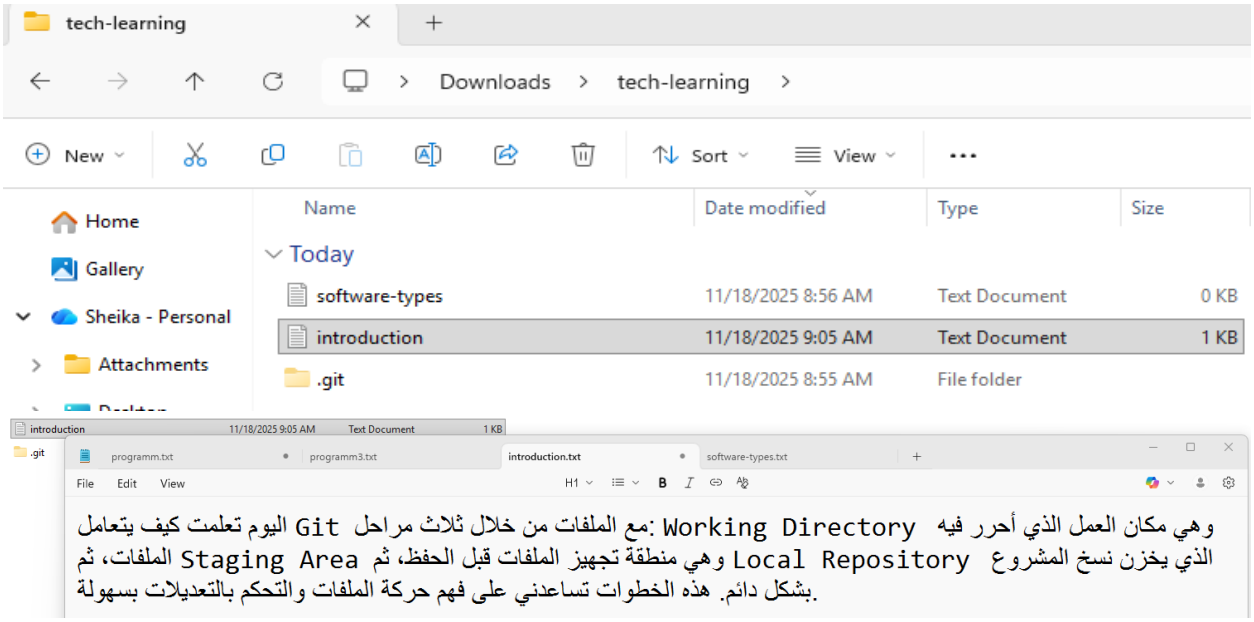
- Create a new folder called tech-learning



- Initialize it as a Git repository

```
Microsoft Windows [Version 10.0.26200.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sheik\Downloads\tech-learning>git init
Initialized empty Git repository in C:/Users/sheik/Downloads/tech-learning/.git/
```

- Create a file introduction.txt and write 3–4 sentences about what you learned today about software



- Check the status

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        introduction.txt

nothing added to commit but untracked files present (use "git add" to track)
```

- Add the file to staging area

```
C:\Users\sheik\Downloads\tech-learning>git add .
```

- Commit with message "Add introduction to tech journal"

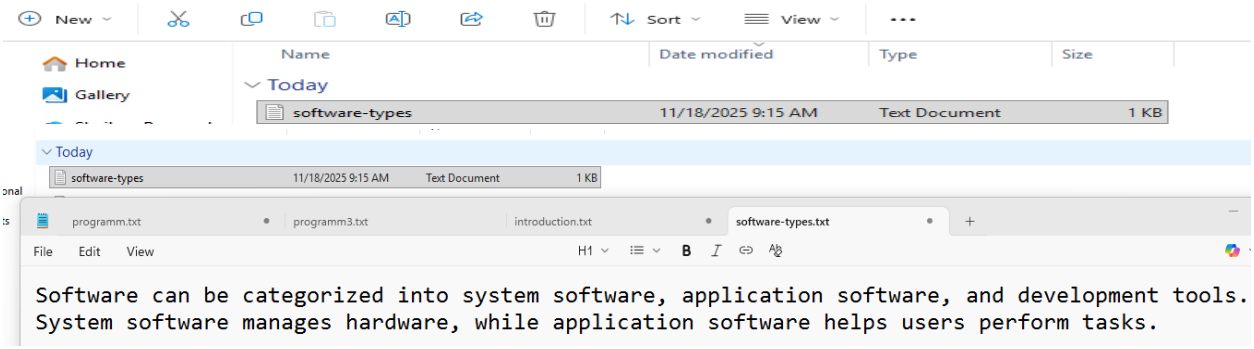
```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add introduction to tech journal"
[master (root-commit) beb0c8f] Add introduction to tech journal
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 introduction.txt
```

- View your commit log in one-line format

```
C:\Users\sheik\Downloads\tech-learning>git log --oneline
beb0c8f (HEAD -> master) Add introduction to tech journal

C:\Users\sheik\Downloads\tech-learning>
```

Task 1.2: Software Types Overview

- Create a file software-types.txt and write 2–3 lines about software types



- Check status (untracked file)

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   introduction.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        software-types.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

- Add and commit with message "Add software types overview"

```
:\Users\sheik\Downloads\tech-learning>git add software-types.txt

:\Users\sheik\Downloads\tech-learning>git commit -m "Add software types overview"
master a8dcb8b] Add software types overview
1 file changed, 2 insertions(+)
create mode 100644 software-types.txt
```

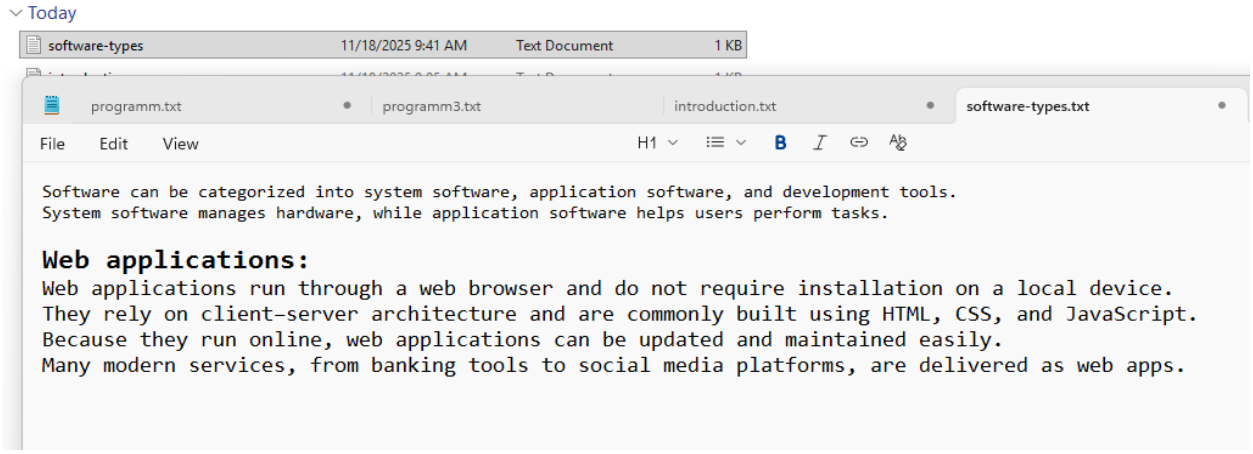
- View log – should see 2 commits

```
C:\Users\sheik\Downloads\tech-learning>git log --oneline
a8dcb8b (HEAD -> master) Add software types overview
beb0c8f Add introduction to tech journal

C:\Users\sheik\Downloads\tech-learning>git add .
```

Level 2: Building Content Gradually

Task 2.1: Expanding Software Types

- Add a section about Web Applications (3–4 sentences)



- Check status, add, commit with "Add web applications section"

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working director
y)
        modified:   introduction.txt
        modified:   software-types.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\sheik\Downloads\tech-learning>git add .

C:\Users\sheik\Downloads\tech-learning>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   introduction.txt
        modified:   software-types.txt


C:\Users\sheik\Downloads\tech-learning>git commit -m "Add web applica
tions section"
[master 2f6308e] Add web applications section
 2 files changed, 8 insertions(+)
```

- View log – 3 commits

```
C:\Users\sheik\Downloads\tech-learning>git log --oneline
2f6308e (HEAD -> master) Add web applications section
a8dcb8b Add software types overview
beb0c8f Add introduction to tech journal

C:\Users\sheik\Downloads\tech-learning>
```

Task 2.2: Adding More Software Types

- Add Mobile Applications section (3–4 sentences), add & commit

Today

software-types 11/18/2025 10:03 AM Text Document 1 KB

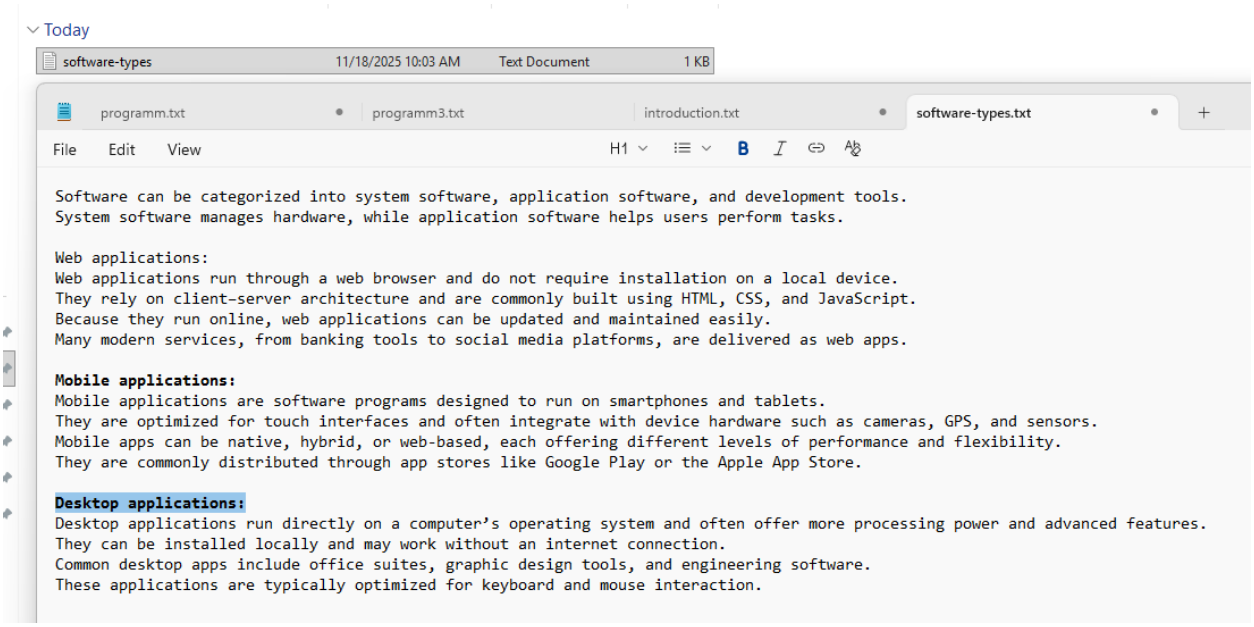
programm.txt | programm3.txt | introduction.txt | software-types.txt

File Edit View H1

```
Software can be categorized into system software, application software, and development tools.
System software manages hardware, while application software helps users perform tasks.

Web applications:
Web applications run through a web browser and do not require installation on a local device.
They rely on client-server architecture and are commonly built using HTML, CSS, and JavaScript.
Because they run online, web applications can be updated and maintained easily.
Many modern services, from banking tools to social media platforms, are delivered as web apps.

Mobile applications:
Mobile applications are software programs designed to run on smartphones and tablets.
They are optimized for touch interfaces and often integrate with device hardware such as cameras, GPS, and sensors.
Mobile apps can be native, hybrid, or web-based, each offering different levels of performance and flexibility.
They are commonly distributed through app stores like Google Play or the Apple App Store.
```

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working director
y)
        modified:   software-types.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\sheik\Downloads\tech-learning>git add .

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add Mobile appl
ications section"
[master f14aafc] Add Mobile applications section
 1 file changed, 8 insertions(+), 1 deletion(-)

C:\Users\sheik\Downloads\tech-learning>git log --oneline
f14aafc (HEAD -> master) Add Mobile applications section
2f6308e Add web applications section
a8dcb8b Add software types overview
beb0c8f Add introduction to tech journal

C:\Users\sheik\Downloads\tech-learning>
```

- Add Desktop Applications section, add & commit



```
C:\Users\sheik\Downloads\tech-learning>git add software-types.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add desktop app
lications section"
[master cc26e13] Add desktop applications section
 1 file changed, 7 insertions(+)
```

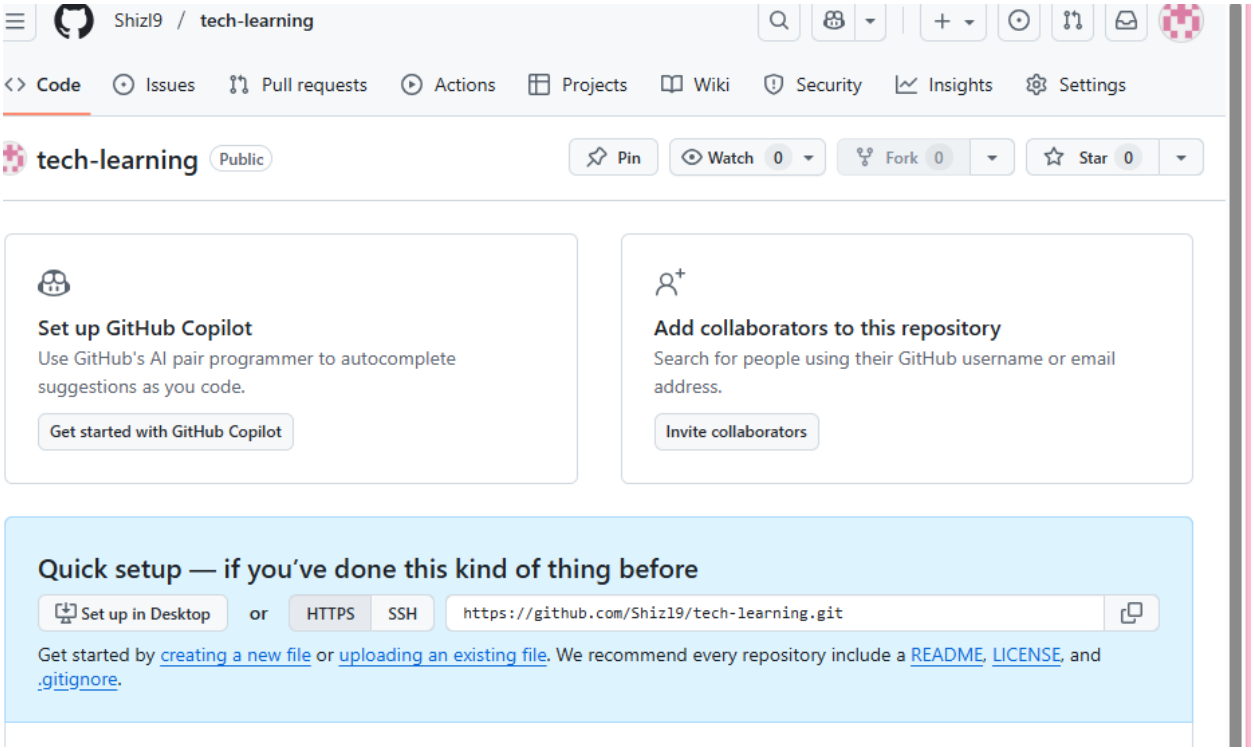
- View log – 5 commits

```
C:\Users\sheik\Downloads\tech-learning>git log --oneline
cc26e13 (HEAD -> master) Add desktop applications section
f14aafc Add Mobile applications section
2f6308e Add web applications section
a8dcb8b Add software types overview
beb0c8f Add introduction to tech journal

C:\Users\sheik\Downloads\tech-learning>
```

Level 3: Connect to GitHub

Task 3.1: Publishing Your Work

- Create GitHub repo tech-learning



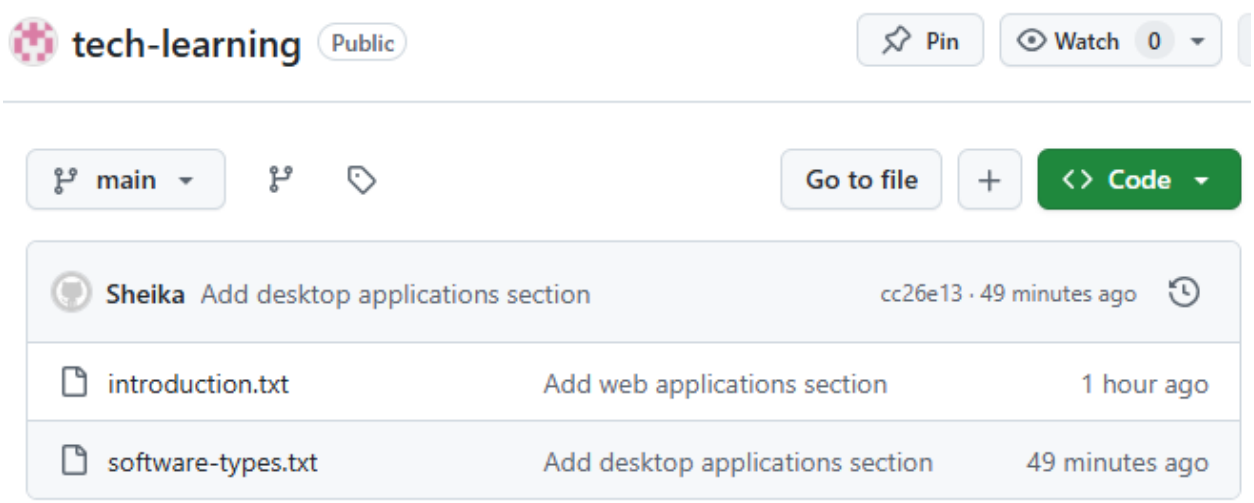
- Add remote origin

```
C:\Users\sheik\Downloads\tech-learning>git remote add origin https://
github.com/Shizl9/tech-learning.git
```

- Push commits

```
C:\Users\sheik\Downloads\tech-learning>git branch -M main

C:\Users\sheik\Downloads\tech-learning>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 12 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (16/16), 2.20 KiB | 1.10 MiB/s, done.
Total 16 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/Shizl9/tech-learning.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\sheik\Downloads\tech-learning>
```

- Verify files and commits on GitHub



Shizl9 / tech-learning

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

main | tech-learning / introduction.txt | Go to file

Sheika Add web applications section 2f6308e · 1 hour ago

1 lines (1 loc) · 475 Bytes

Code | Blame | Raw

```
1  اليوم تعلمت كيف يتعامل Git مراحل ثلاث خلال من الملفات مع: Working Directory ثم الملفات، فيه أحرر الذي العمل مكان وهي Staging Area ثم الحفظ قبل الملفات جهيز
```

main | tech-learning / software-types.txt | Go to file

Sheika Add desktop applications section cc26e13 · 51 minutes ago

3 lines (17 loc) · 1.36 KB

Code | Blame | Raw

```
1  Software can be categorized into system software, application software, and development tools.
2  System software manages hardware, while application software helps users perform tasks.
3
4  Web applications:
5  Web applications run through a web browser and do not require installation on a local device.
6  They rely on client-server architecture and are commonly built using HTML, CSS, and JavaScript.
7  Because they run online, web applications can be updated and maintained easily.
8  Many modern services, from banking tools to social media platforms, are delivered as web apps.
9
10 Mobile applications:
11 Mobile applications are software programs designed to run on smartphones and tablets.
12 They are optimized for touch interfaces and often integrate with device hardware such as cameras, GPS, and sensors.
13 Mobile apps can be native, hybrid, or web-based, each offering different levels of performance and flexibility.
14 They are commonly distributed through app stores like Google Play or the Apple App Store.
15
16 Desktop applications:
17 Desktop applications run directly on a computer's operating system and often offer more processing power and advanced featu
18 They can be installed locally and may work without an internet connection.
19 Common desktop apps include office suites, graphic design tools, and engineering software.
20 These applications are typically optimized for keyboard and mouse interaction.
```

Task 3.2: Continuing Work After Push

- Create my-learning-notes.txt

Today

my-learning-notes	11/18/2025 11:22 AM	Text Document	1 KB

- Write what you learned about Git commands

Today

my-learning-notes 11/18/2025 11:22 AM Text Document 1 KB

programm.txt programm3.txt introduction.txt software-types.txt

File Edit View H1

```
I learned how to use Git commands to track changes, commit files, and push to GitHub.
I also learned about branches, checking status, and viewing logs.
```

- Add, commit, push, verify on GitHub

```
C:\Users\sheik\Downloads\tech-learning>git add my-learning-notes.txt
```

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add my learning
 notes about Git commands"
[main 20eb4bc] Add my learning notes about Git commands
 1 file changed, 2 insertions(+)
 create mode 100644 my-learning-notes.txt
```

```
C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 472 bytes | 472.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Shizl9/tech-learning.git
   cc26e13..20eb4bc  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

main Go to file + Code

Sheika Add my learning notes about Git commands 20eb4bc · 4 minutes ago

introduction.txt	Add web applications section	1 hour ago
my-learning-notes.txt	Add my learning notes about Git c...	4 minutes ago
software-types.txt	Add desktop applications section	1 hour ago

main tech-learning / my-learning-notes.txt Go to file t

Sheika Add my learning notes about Git commands 20eb4bc · 5 minutes ago

2 lines (2 loc) · 152 Bytes

Code Blame Raw

```
I learned how to use Git commands to track changes, commit files, and push to GitHub.
I also learned about branches, checking status, and viewing logs.
```

Level 4: Web Development Basics

Task 4.1: Web Development Parts

- Create web-development.txt, write title & intro

Today

web-development 11/18/2025 11:43 AM Text Document 1 KB

programm.txt programm3.txt introduction.txt software-types.txt my-learning-no

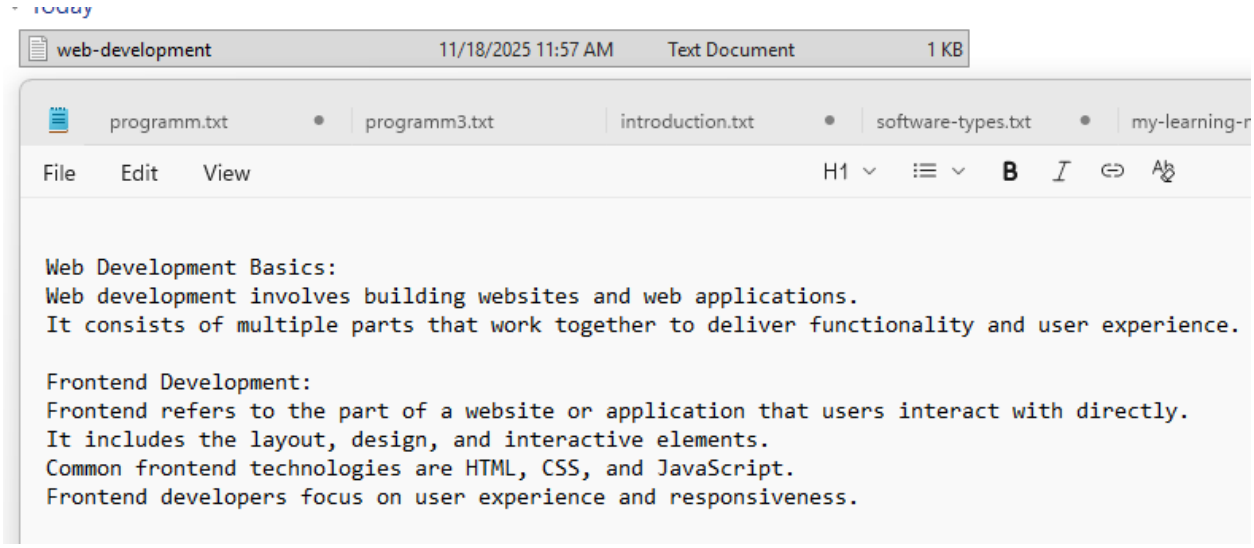
File Edit View H1

```
Web Development Basics
Web development involves building websites and web applications.
It consists of multiple parts that work together to deliver functionality and user experience.
```

- Commit "Add web development introduction"

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add web develop
ment introduction"
[main 1a016c1] Add web development introduction
 1 file changed, 4 insertions(+)
 create mode 100644 web-development.txt

C:\Users\sheik\Downloads\tech-learning>
```

- Add Frontend section (4–5 sentences), commit




```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add frontend se
ction"
[main 32b377f] Add frontend section
 1 file changed, 8 insertions(+), 2 deletions(-)

C:\Users\sheik\Downloads\tech-learning>
```

- Add Backend section, commit

Today

web-development	11/18/2025 12:03 PM	Text Document	1 KB

programm.txt • programm3.txt introduction.txt • software-types.txt • my-learning-notes.txt web-dev

File Edit View H1

```
Web Development Basics:
Web development involves building websites and web applications.
It consists of multiple parts that work together to deliver functionality and user experience.

Frontend Development:
Frontend refers to the part of a website or application that users interact with directly.
It includes the layout, design, and interactive elements.
Common frontend technologies are HTML, CSS, and JavaScript.
Frontend developers focus on user experience and responsiveness.

Backend Development:
Backend is the server-side part of a web application that handles logic, databases, and server communication.
It ensures that the frontend receives the correct data and functionality.
Common backend technologies include Python, Node.js, Java, and PHP.
Backend developers focus on performance, security, and scalability.
```

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add backend sec
tion"
[main 66c8252] Add backend section
 1 file changed, 6 insertions(+)

C:\Users\sheik\Downloads\tech-learning>
```

- Add Database section, commit

Today

web-development 11/18/2025 12:06 PM Text Document 2 KB

programm.txt | programm3.txt | introduction.txt | software-types.txt | my-learning-notes.txt | web-developr

File Edit View H1

```
Web Development Basics:
Web development involves building websites and web applications.
It consists of multiple parts that work together to deliver functionality and user experience.

Frontend Development:
Frontend refers to the part of a website or application that users interact with directly.
It includes the layout, design, and interactive elements.
Common frontend technologies are HTML, CSS, and JavaScript.
Frontend developers focus on user experience and responsiveness.

Backend Development:
Backend is the server-side part of a web application that handles logic, databases, and server communication.
It ensures that the frontend receives the correct data and functionality.
Common backend technologies include Python, Node.js, Java, and PHP.
Backend developers focus on performance, security, and scalability.

Databases:
Databases store and organize the data used by web applications.
They can be relational (like MySQL, PostgreSQL) or non-relational (like MongoDB).
Databases allow efficient retrieval, update, and management of data.
Backend systems communicate with databases to provide information to the frontend.
```

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add database se
ction"
[main 0510d79] Add database section
 1 file changed, 6 insertions(+)

C:\Users\sheik\Downloads\tech-learning>
```

- Push to GitHub

```
C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.49 KiB | 1.49 MiB/s, done.
Total 12 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 1 local object.
To https://github.com/Shizl9/tech-learning.git
   20eb4bc..0510d79  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

tech-learning Public

Pin | Watch 0 | F

main | Go to file | + | Code | Ab

Sheika Add database section 0510d79 · 2 minutes ago

introduction.txt	Add web applications section	2 hours ago
my-learning-notes.txt	Add my learning notes about Git c...	39 minutes ago
software-types.txt	Add desktop applications section	1 hour ago
web-development.txt	Add database section	2 minutes ago

Re

main | tech-learning / web-development.txt | Go to file t

Sheika Add database section 0510d79 · 3 minutes ago

22 lines (18 loc) · 1.12 KB

Code | Blame | Raw

```

Web Development Basics:
Web development involves building websites and web applications.
It consists of multiple parts that work together to deliver functionality and user experience.

Frontend Development:
Frontend refers to the part of a website or application that users interact with directly.
It includes the layout, design, and interactive elements.
Common frontend technologies are HTML, CSS, and JavaScript.
Frontend developers focus on user experience and responsiveness.

Backend Development:
Backend is the server-side part of a web application that handles logic, databases, and server communication.
It ensures that the frontend receives the correct data and functionality.
Common backend technologies include Python, Node.js, Java, and PHP.
Backend developers focus on performance, security, and scalability.

Databases:
Databases store and organize the data used by web applications.
They can be relational (like MySQL, PostgreSQL) or non-relational (like MongoDB).
Databases allow efficient retrieval, update, and management of data.
Backend systems communicate with databases to provide information to the frontend.
```

Task 4.2: Remote Edit Practice

- Edit web-development.txt on GitHub, add conclusion, commit

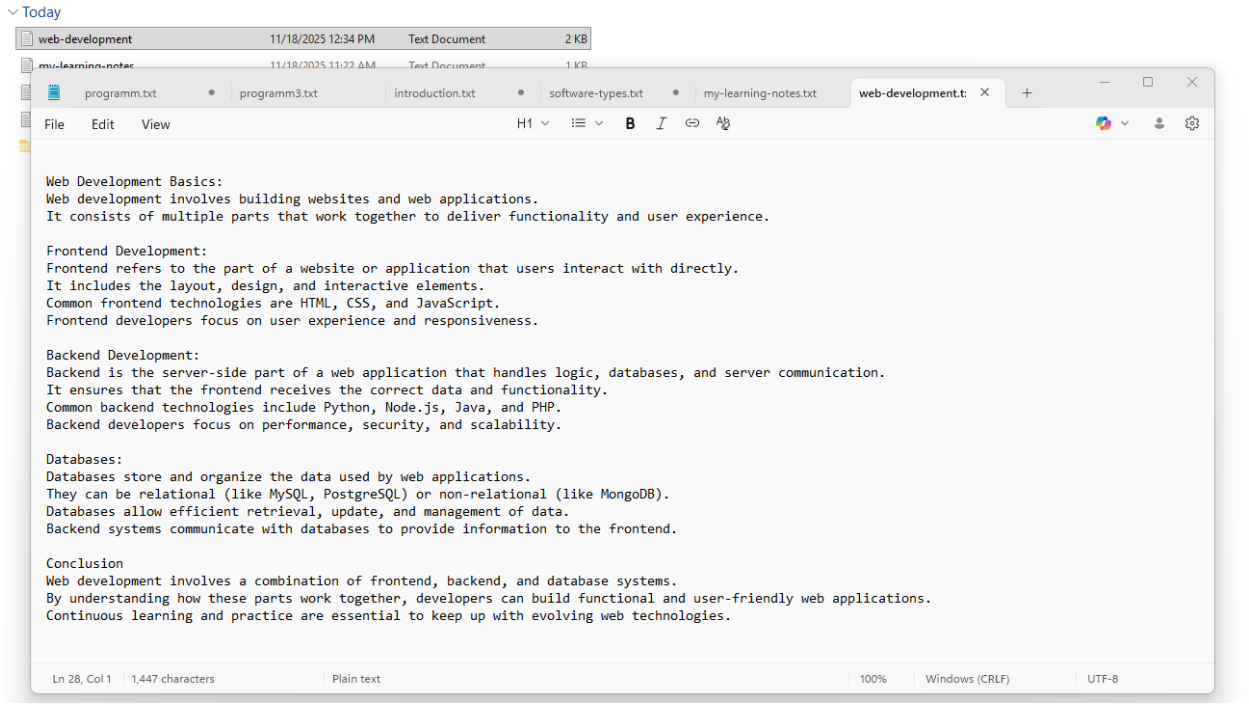
Edit | Preview | Spaces | 2 | Soft wrap

```

Web Development Basics:
Web development involves building websites and web applications.
It consists of multiple parts that work together to deliver functionality and user experience.

Frontend Development:
Frontend refers to the part of a website or application that users interact with directly.
It includes the layout, design, and interactive elements.
Common frontend technologies are HTML, CSS, and JavaScript.
Frontend developers focus on user experience and responsiveness.

Backend Development:
Backend is the server-side part of a web application that handles logic, databases, and server communication.
It ensures that the frontend receives the correct data and functionality.
Common backend technologies include Python, Node.js, Java, and PHP.
Backend developers focus on performance, security, and scalability.

Databases:
Databases store and organize the data used by web applications.
They can be relational (like MySQL, PostgreSQL) or non-relational (like MongoDB).
Databases allow efficient retrieval, update, and management of data.
Backend systems communicate with databases to provide information to the frontend.

Conclusion:
Web development involves a combination of frontend, backend, and database systems.
By understanding how these parts work together, developers can build functional and user-friendly web applications.
Continuous learning and practice are essential to keep up with evolving web technologies.
```

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add conclusion
to web development notes"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

- Pull changes locally and verify

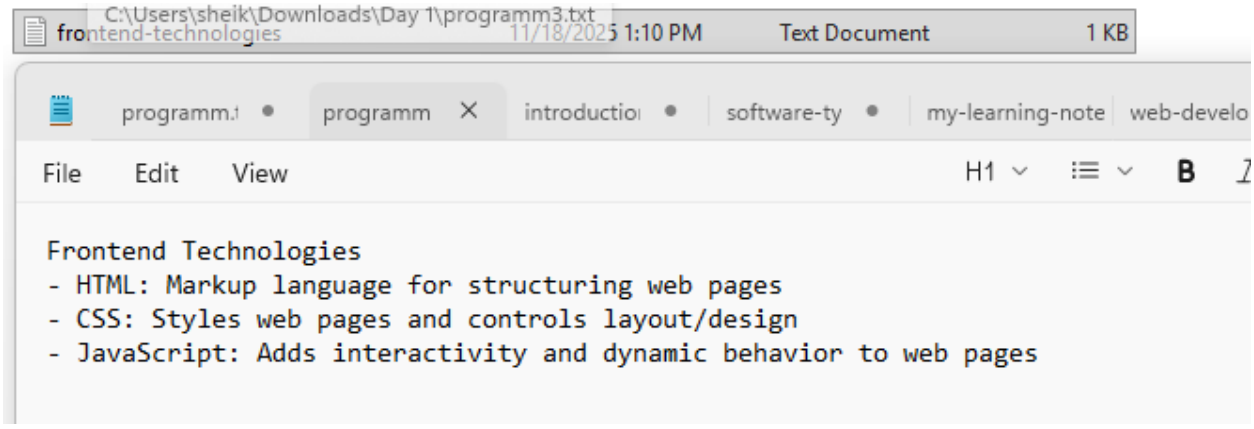
```
C:\Users\sheik\Downloads\tech-learning>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0
Unpacking objects: 100% (3/3), 1.12 KiB | 287.00 KiB/s, done.
From https://github.com/Shizl9/tech-learning
   0510d79..e6aeb0a  main       -> origin/main
Updating 0510d79..e6aeb0a
Fast-forward
 web-development.txt | 5 +++++
 1 file changed, 5 insertions(+)

C:\Users\sheik\Downloads\tech-learning>
```



Level 5: Multiple Files Project

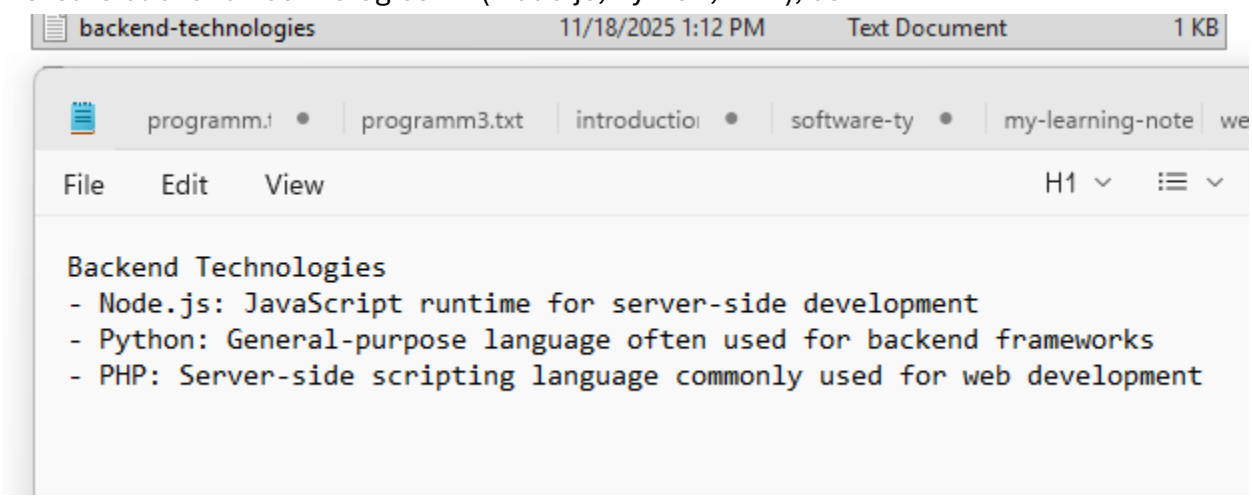
Task 5.1: Technology Stacks Research

- Create frontend-technologies.txt (HTML, CSS, JS), commit




```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add frontend te
chnologies file"
[main 8218362] Add frontend technologies file
 1 file changed, 4 insertions(+)
 create mode 100644 frontend-technologies.txt
```

- Create backend-technologies.txt (Node.js, Python, PHP), commit




```
C:\Users\sheik\Downloads\tech-learning>git add backend-technologies.t
xt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add backend tec
hnologies file"
[main 3e2d016] Add backend technologies file
 1 file changed, 4 insertions(+)
 create mode 100644 backend-technologies.txt
```

- Create database-technologies.txt (MySQL, MongoDB, PostgreSQL), commit

✓ Today

database-technologies 11/18/2025 1:15 PM Text Document 1 KB

programm.t • programm3.txt introductio • software-ty • my-learning-note web-developmen frontend-te

File Edit View H1

```
Database Technologies
- MySQL: Relational database management system
- MongoDB: NoSQL document-oriented database
- PostgreSQL: Advanced open-source relational database
```

```
C:\Users\sheik\Downloads\tech-learning>git commit -m "Add database te
chnologies file"
[main a2b0188] Add database technologies file
 1 file changed, 4 insertions(+)
 create mode 100644 database-technologies.txt
```

- Push commits

```
C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.29 KiB | 1.29 MiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Shizl9/tech-learning.git
   e6aeb0a..a2b0188  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

main | Go to file | + | Code

Sheika Add database technologies file a2b0188 · 10 minutes ago

backend-technologies.txt	Add backend technologies file	11 minutes ago
database-technologies.txt	Add database technologies file	10 minutes ago
frontend-technologies.txt	Add frontend technologies file	14 minutes ago
introduction.txt	Add web applications section	3 hours ago
my-learning-notes.txt	Add my learning notes about Git ...	2 hours ago
software-types.txt	Add desktop applications section	3 hours ago
web-development.txt	Update web-development.txt	1 hour ago

Task 5.2: Updating Multiple Files

- Modify frontend-technologies.txt (React/Vue), save

```
Frontend Technologies
- HTML: Markup language for structuring web pages
- CSS: Styles web pages and controls layout/design
- JavaScript: Adds interactivity and dynamic behavior to web pages
- React: JavaScript library for building user interfaces
- Vue: Progressive JavaScript framework for building interfaces
```

- Modify backend-technologies.txt (Java/Ruby), save

programm | programm3.txt | introductio | software-ty | my-learning-note | web-dev

File Edit View H1 B

```
Backend Technologies
- Node.js: JavaScript runtime for server-side development
- Python: General-purpose language often used for backend frameworks
- PHP: Server-side scripting language commonly used for web development
- Java: Popular language for enterprise backend applications
- Ruby: Dynamic language often used with Rails framework
```

- Stage ONLY frontend file, commit

```
C:\Users\sheik\Downloads\tech-learning>git add frontend-technologies.
txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Update frontend
 technologies with React and Vue"
[main 5f74890] Update frontend technologies with React and Vue
 1 file changed, 3 insertions(+), 1 deletion(-)
```

- Stage backend file, commit

```
C:\Users\sheik\Downloads\tech-learning>git add backend-technologies.t
xt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Update backend
technologies with Java and Ruby"
[main 4187a36] Update backend technologies with Java and Ruby
 1 file changed, 3 insertions(+)
```

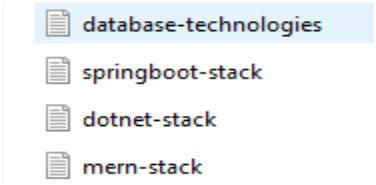
- Push

```
C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 902 bytes | 902.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects
To https://github.com/Shizl9/tech-learning.git
   a2b0188..4187a36  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

Level 6: Development Stacks Deep Dive

Task 6.1: Popular Stacks Documentation

- Create mern-stack.txt, dotnet-stack.txt, springboot-stack.txt



```
C:\Users\sheik\Downloads\tech-learning>git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        mern-stack.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\sheik\Downloads\tech-learning>git add mern-stack.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add MERN stack documentation"
[main 241de4a] Add MERN stack documentation
 1 file changed, 4 insertions(+)
 create mode 100644 mern-stack.txt
```

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        dotnet-stack.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\sheik\Downloads\tech-learning>git add dotnet-stack.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add DotNet stack documentation"
[main e18f751] Add DotNet stack documentation
 1 file changed, 4 insertions(+)
 create mode 100644 dotnet-stack.txt
```

```
C:\Users\sheik\Downloads\tech-learning>git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        springboot-stack.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\sheik\Downloads\tech-learning>git add springboot-stack.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add Spring Boot stack documentation"
[main 21148db] Add Spring Boot stack documentation
 1 file changed, 4 insertions(+)
 create mode 100644 springboot-stack.txt
```

- Explain stack components (frontend, backend, DB)

mern-stack.txt | dotnet-stack.txt

File Edit View

```
MERN Stack
Frontend: React.js for building dynamic user interfaces.
Backend: Node.js with Express.js for server-side logic.
Database: MongoDB for NoSQL document storage.
```

mern-stack.txt | dotnet-stack.txt | springb

File Edit View H1

```
DotNet Stack
Frontend: Blazor, Razor Pages, or external frontend frameworks.
Backend: ASP.NET Core for building powerful server-side applications.
Database: SQL Server or PostgreSQL commonly used.
```

mern-stack.txt | dotnet-stack.txt | spri

File Edit View H1

```
Spring Boot Stack
Frontend: Can use React, Angular, or Thymeleaf for UI.
Backend: Java Spring Boot for REST APIs and microservices.
Database: MySQL, PostgreSQL, or MongoDB.
```

- Push commits

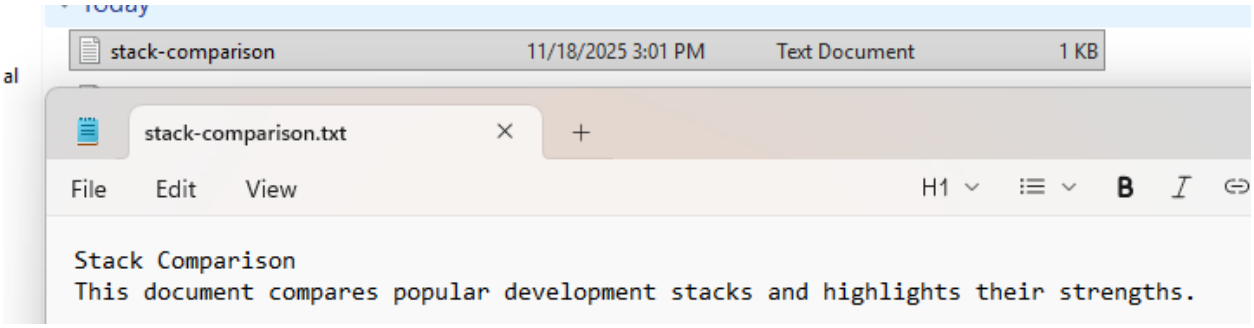
```
C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.19 KiB | 1.19 MiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/Shizl9/tech-learning.git
   4187a36..21148db  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

Sheika Add Spring Boot stack documentation 21148db · 10 minutes ago

backend-technologies.txt	Update backend technologies wit...	24 minutes ago
database-technologies.txt	Add database technologies file	1 hour ago
dotnet-stack.txt	Add DotNet stack documentation	11 minutes ago
frontend-technologies.txt	Update frontend technologies wit...	25 minutes ago
introduction.txt	Add web applications section	5 hours ago
mern-stack.txt	Add MERN stack documentation	14 minutes ago
my-learning-notes.txt	Add my learning notes about Git ...	3 hours ago
software-types.txt	Add desktop applications section	4 hours ago
springboot-stack.txt	Add Spring Boot stack document...	10 minutes ago
web-development.txt	Update web-development.txt	2 hours ago

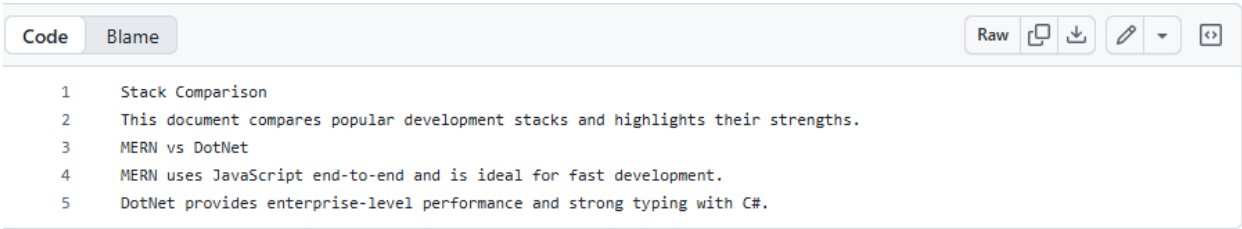
Task 6.2: Comparison Document

- Create stack-comparison.txt with intro, commit, push




```
no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\sheik\Downloads\tech-learning>git add stack-comparison.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add stack compa
rison introduction"
[main 8a9cc00] Add stack comparison introduction
 1 file changed, 2 insertions(+)
 create mode 100644 stack-comparison.txt

C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 389 bytes | 389.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Shizl9/tech-learning.git
   21148db..8a9cc00  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

- Edit on GitHub: add MERN vs DotNet comparison



- Locally add MERN vs Spring Boot comparison (without pulling)

```
Stack Comparison
This document compares popular development stacks and highlights their strengths.
MERN vs Spring Boot
MERN is lightweight and JavaScript-based.
Spring Boot is powerful, scalable, and uses Java for enterprise systems.
```

```
C:\Users\sheik\Downloads\tech-learning>git add stack-comparison.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add MERN vs Spr
ing Boot comparison"
[main 94f0cbd] Add MERN vs Spring Boot comparison
 1 file changed, 3 insertions(+), 3 deletions(-)
```

- Attempt push → conflict

```
C:\Users\sheik\Downloads\tech-learning>git push
To https://github.com/Shizl9/tech-learning.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/Shizl9/tech-le
arning.git'
hint: Updates were rejected because the remote contains work that you
 do not
hint: have locally. This is usually caused by another repository push
ing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for det
ails.
```

- Pull, resolve, verify both comparisons

```
C:\Users\sheik\Downloads\tech-learning>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0
Unpacking objects: 100% (3/3), 1.06 KiB | 155.00 KiB/s, done.
From https://github.com/Shizl9/tech-learning
   8a9cc00..5cd0c8b  main       -> origin/main
Updating 8a9cc00..5cd0c8b
Fast-forward
 stack-comparison.txt | 5 ++++-
 1 file changed, 4 insertions(+), 1 deletion(-)

C:\Users\sheik\Downloads\tech-learning>
```

- Add conclusion locally, commit, push

stack-comparison.txt

File Edit View H1

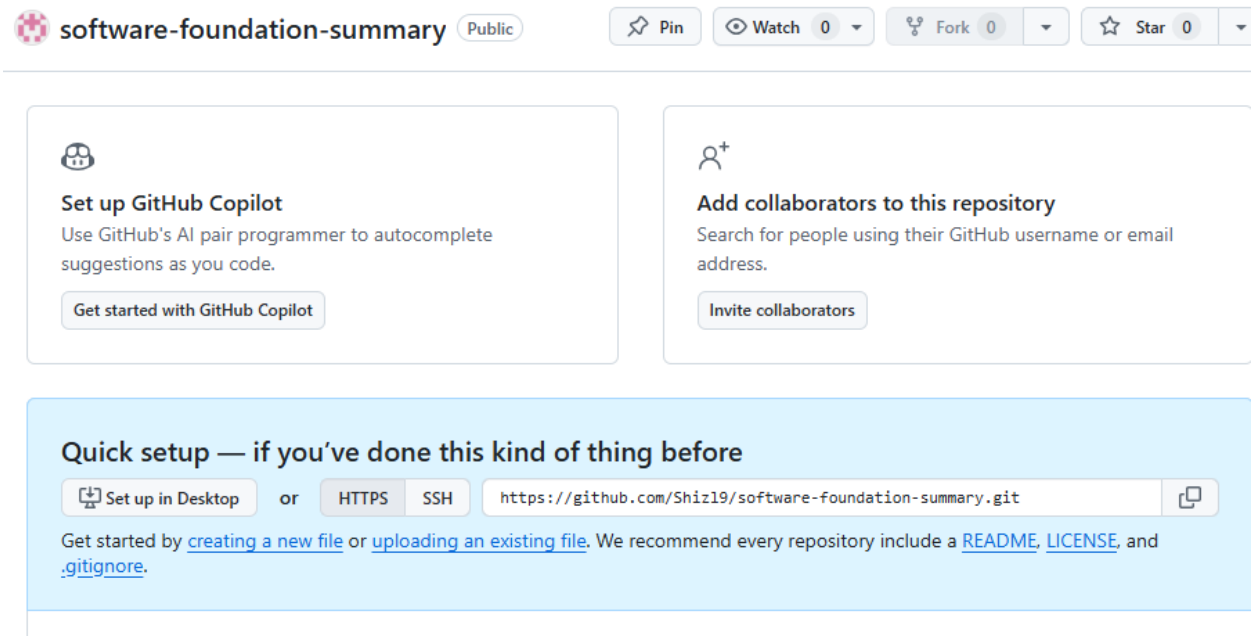
```
Stack Comparison
This document compares popular development stacks and highlights their strengths.
MERN vs Spring Boot
MERN is lightweight and JavaScript-based.
Spring Boot is powerful, scalable, and uses Java for enterprise systems.
Conclusion
Each stack has unique advantages, and the best choice depends on project requirements, scalability needs, and developer expertise.
```

```
C:\Users\sheik\Downloads\tech-learning>git add stack-comparison.txt

C:\Users\sheik\Downloads\tech-learning>git commit -m "Add conclusion
to stack comparison"
[main c43da56] Add conclusion to stack comparison
 1 file changed, 3 insertions(+)

C:\Users\sheik\Downloads\tech-learning>git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 427 bytes | 427.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects
To https://github.com/Shizl9/tech-learning.git
   94f0cbd..c43da56  main -> main

C:\Users\sheik\Downloads\tech-learning>
```

Compound task part 2 (Only proceed if you have finished part 1 (6 levels above))

Level 7

1- Make a new GitHub repository name (software foundation summary)



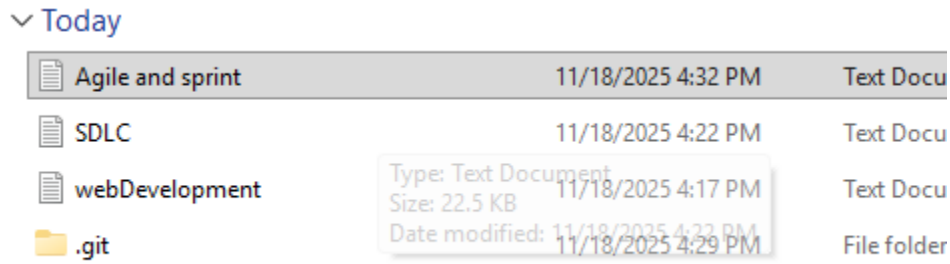
2- On your local computer, make a new folder(software foundation summary) and Initialize git in it

```
C:\Users\sheik\Downloads\software foundation summary>git init
Initialized empty Git repository in C:/Users/sheik/Downloads/software foundation summary/.git/

C:\Users\sheik\Downloads\software foundation summary>
```

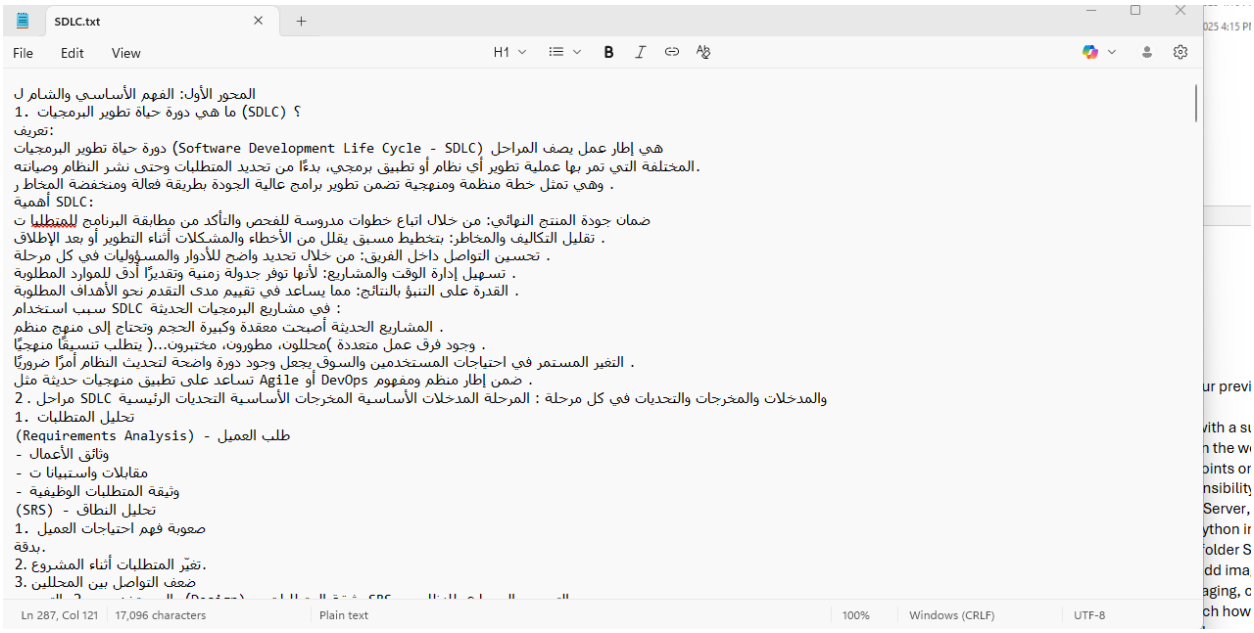
3- Connect the remote repo with your local folder (hint: remote add command)

```
C:\Users\sheik\Downloads\software foundation summary>git remote add origin https://github.com/Shizl9/s
oftware-foundation-summary.git

C:\Users\sheik\Downloads\software foundation summary>
```

4- In your local folder, add 3 text files (SDLC.txt, Agile_and_sprint.txt, webDevelopment.txt



5- Open the SDLC.txt and copy & paste your previous research report about SDLC in here



SDLC.txt

File Edit View

المحور الأول: الفهم الأساسي والشام ل
1. ما هي دورة حياة تطوير البرمجيات (SDLC) ؟
تعريف:
هي إطار عمل يصف المراحل (Software Development Life Cycle - SDLC) دورة حياة تطوير البرمجيات
المختلفة التي تمر بها عملية تطوير أي نظام أو تطبيق برمجي، بدءًا من تحديد المتطلبات وحتى نشر النظام وصيانته.
وهي تمثل خطة منظمة ومنهجية تضمن تطوير برامج عالية الجودة بطريقة فعالة ومنخفضة المخاط ر .
أهمية SDLC:
ضمان جودة المنتج النهائي: من خلال اتباع خطوات مدروسة للفحص والتأكد من مطابقة البرنامج للمتطلبا ت
تقليل التكاليف والمخاطر: بتخطيط مسبق يقلل من الأخطاء والمشكلات أثناء التطوير أو بعد الإطلاق .
تحسين التواصل داخل الفريق: من خلال تحديد واضح للأدوار والمسؤوليات في كل مرحلة .
تسهيل إدارة الوقت والمشاريع: لأنها توفر جدولة زمنية وتقديرًا أدق للموارد المطلوبة .
القدرة على التنبؤ بالنتائج: مما يساعد في تقييم مدى التقدم نحو الأهداف المطلوبة .
سبب استخدام SDLC في مشاريع البرمجيات الحديثة :
المشاريع الحديثة أصبحت معقدة وكبيرة الحجم وتحتاج إلى منهج منظم .
وجود فرق عمل متعددة)محللون، مطورون، مختبرون...(يتطلب تنسيقًا منهجيًا .
التغير المستمر في احتياجات المستخدمين والسوق يجعل وجود دورة واضحة لتحديث النظام أمرًا ضروريًا .
ضمن إطار منظم ومفهوم DevOps أو Agile تساعد على تطبيق منهجيات حديثة مثل .
2 . مراحل SDLC الرئيسية التحديات الأساسية المخرجات الأساسية المرحلة : المدخلات والمخرجات والتحديات في كل مرحلة
1. تحليل المتطلبات
(Requirements Analysis) - طلب العميل
- وثائق الأعمال
- مقابلات واستبيانا ت
- وثيقة المتطلبات الوظيفية
(SRS) - تحليل النطاق
1. صعوبة فهم احتياجات العميل
بدقة.
2. تغيّر المتطلبات أثناء المشروع.
3. ضعف التواصل بين المحللين

Ln 287, Col 121 | 17,096 characters | Plain text | 100% | Windows (CRLF) | UTF-8

6- save, add to the staging area, commit with a suitable message and push

```
C:\Users\sheik\Downloads\software foundation summary>git add SDLC.txt

C:\Users\sheik\Downloads\software foundation summary>git commit -m "Add SDLC report"
[master (root-commit) 0a4d5b2] Add SDLC report
 1 file changed, 287 insertions(+)
 create mode 100644 SDLC.txt

C:\Users\sheik\Downloads\software foundation summary>git push
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin master

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.


C:\Users\sheik\Downloads\software foundation summary>
```

7- Repeat steps 5,6 with the Agile_and_sprint.txt

Agile and sprint.txt

File Edit View

مجافظة الفريق على الانسجام والتركيز على أهداف ال Sprint.
•
مشاركة كل عضوٍ ما تم إنجازه بالأمس، وما سيعمل عليه اليوم، وما هي العوائق .
•
الكشف المبكر عن المشاكل وحلها قبل أن تؤثر على سير العمل .
•
تحسين التواصل والتعاون بين أعضاء الفريق .
•
الحفاظ على زخم العمل والتقدم المستمر طوالً فترة ال Sprint.
4. الفرق بين Product Log و Sprint Log
•
Product Log / Product Backlog:
o
قائمة رئيسية تحتوي على جميع الميزات، التحسينات، الأخطاء، والمتطلبات للمنتج .
o
وتُحدّث باستمرار حسب الأولويات والتغيرات Product Owner يملكها .
o
تمثل نطاق المنتج الكامل .
•
Sprint Log / Sprint Backlog:
o
مجموعة مختارة منً ال Product Backlog سيتم تنفيذها خلال Sprint معين .
o
يملكها فريق التطوير وتوضح خطة إنجاز المهام اليومية .
o
الحالي Sprint تمثل ما سيتم إنجازه خلال .
باختصار: Product Backlog يحدد ما يمكن فعلهً بينما Sprint Backlog يحدد ما سيتم فعله خلال
الحالي Sprint .

```
C:\Users\sheik\Downloads\software foundation summary>git add Agile and sprint.txt
fatal: pathspec 'Agile' did not match any files

C:\Users\sheik\Downloads\software foundation summary>git commit -m "Add Agile and Sprint report"
[master 40765d8] Add Agile and Sprint report
 1 file changed, 506 insertions(+)
 create mode 100644 Agile.txt

C:\Users\sheik\Downloads\software foundation summary>
```

8- In the web development file, open it and write a detailed report that covers the following points one by one, each point in a new commit and push after finishing: Explain the responsibility of front-end, backend and database explain in detail the .NET stack tools: SQL Server, C#, EF Core, Linq, Blazor Make a comparison table between C#, Java and Python in terms of compiling, speed, usage, advantages, latest version ... etc 9- In the folder Software Foundation Summary, make a subfolder named (Images), in the folder add images (minimum 4) showing (sdlc, agile, sprint, webdevelopment parts) , add to staging, commit with proper message

webDevelopment.txt

ملف تحرير عرض H1

١) Front-end Responsibilities

- Build the user interface (UI) and user experience (UX).
- Uses HTML, CSS, JavaScript to display content.
- Handles user interactions, buttons, forms, validations.
- Ensures responsive design and fast page performance.

٢) Back-end Responsibilities

- Handles business logic and server operations.
- Processes user requests, authentication, authorization.
- Connects to the database, external APIs, and manages sessions.
- Ensures data security and server performance.

٣) Database Responsibilities

- Stores, retrieves, and manages structured/unstructured data.
- Defines tables, relations, indexes, constraints.
- Ensures data integrity (ACID), backups, performance, and security.

الأسطر ١٧، الأعمدة ١ | ٧١٢ حرفاً | نص عادي | ١٠٠٪ | Windows (CRLI | UTF-8

C:\Windows\System32\cmd.e›

```
C:\Users\sheik\Downloads\software foundation summary>git add
 webDevelopment.txt

C:\Users\sheik\Downloads\software foundation summary>git com
mit -m "Add explanation of frontend, backend, and database r
esponsibilities"
[master (root-commit) 9ce0a4d] Add explanation of frontend,
backend, and database responsibilities
 1 file changed, 16 insertions(+)
 create mode 100644 webDevelopment.txt

C:\Users\sheik\Downloads\software foundation summary>git rem
ote add origin https://github.com/Shizl9/software-foundation
-summary.git

C:\Users\sheik\Downloads\software foundation summary>git rem
ote -v
origin  https://github.com/Shizl9/software-foundation-summar
y.git (fetch)
origin  https://github.com/Shizl9/software-foundation-summar
y.git (push)

C:\Users\sheik\Downloads\software foundation summary>git pus
h
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, u
se

    git push --set-upstream origin master

To have this happen automatically for branches without a tra
cking
upstream, see 'push.autoSetupRemote' in 'git help config'.
```

Shizl9 / software-foundation-summary

Code Issues Pull requests Actions Projects Wiki

Set up GitHub Copilot

Use GitHub's AI pair programmer to autocomplete suggestions as you code.

Explain the responsibility of .NET stack tools: SQL Server, C#, EF Core, LINQ, Blazor

SQL Server: Responsible for storing and managing application data. Acts as the database layer for web applications. Handles data .integrity, queries, and transactions

C#: Main backend programming language in the .NET stack. Used to write server-side logic, handle business rules, and manage .application flow. Object-oriented and scalable

EF Core (Entity Framework Core): ORM tool that allows developers to work with the database using C# objects. Handles CRUD .operations automatically and reduces direct SQL coding

LINQ (Language Integrated Query): Enables querying data (from databases, collections, or XML) directly in C# code. Makes data .manipulation easier and readable. Works with EF Core for database queries

Blazor: Frontend framework for building interactive web applications with C#. Can run in the browser (WebAssembly) or on the .#server. Allows full-stack development using only C

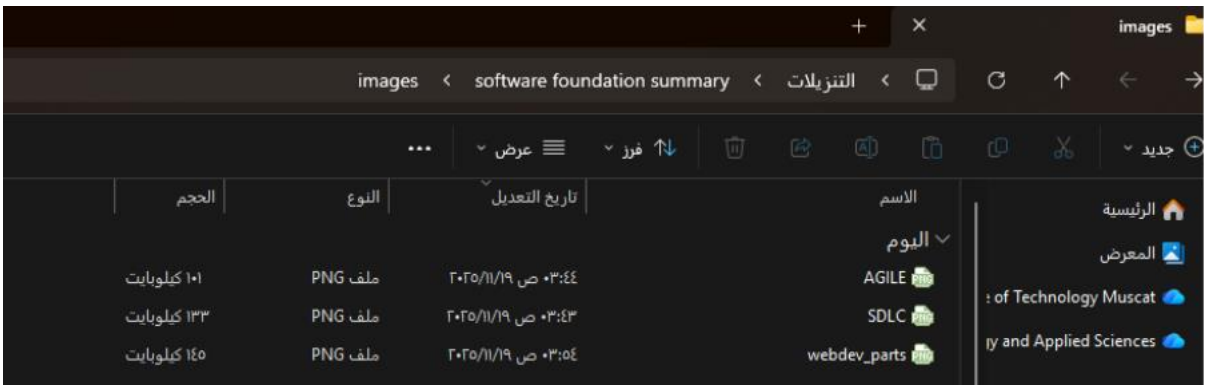
UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٧٠٠ من الأحرف | الأسطر ١٩، الأعمدة ١

```
C:\Users\sheik\Downloads\software foundation summary>git add
 webDevelopment.txt

C:\Users\sheik\Downloads\software foundation summary>git com
mit -m "Add explanation of .NET stack tools: SQL Server, C#,
 EF Core, LINQ, Blazor"
[master 0d12da1] Add explanation of .NET stack tools: SQL Se
rver, C#, EF Core, LINQ, Blazor
 1 file changed, 13 insertions(+)

C:\Users\sheik\Downloads\software foundation summary>
```

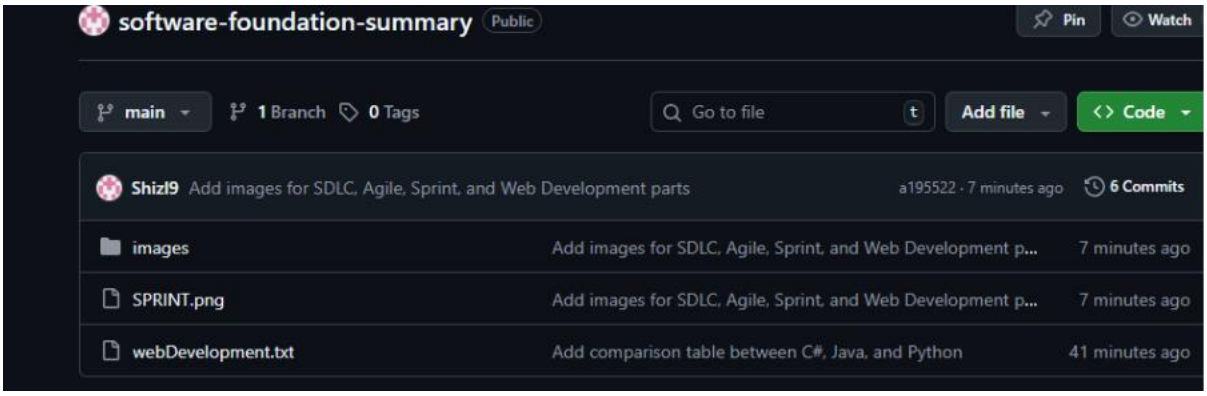
Feature	C#	Java	Python
Compilation**	Compiled to IL (Intermediate Language) then JIT at runtime	Compiled to bytecode, runs on JVM **	Interpreted (can be compiled to bytecode with PyPy)
Speed**	Fast, near Java speed	Fast, platform-independent **	Slower, interpreted
Usage**	Web, desktop, games, enterprise apps	Web, mobile, enterprise apps **	Web, scripting, data science, AI/ML
Advantages**	Strong typing, powerful IDE support, integration with .NET	Platform-independent, mature ecosystem **	Easy syntax, fast development, versatile
Latest Version**	C# 12 (.NET 8)	Java 21 **	Python 3.12
Platform**	Windows, Linux, macOS (.NET Core)	Cross-platform (JVM) **	Cross-platform
Community & Libraries**	Large, especially for Windows apps	Very large, mature **	Huge, especially for AI/ML and web

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ٣,٣٧٦ حرفاً | الأسطر ٤٠، الأعمدة ١

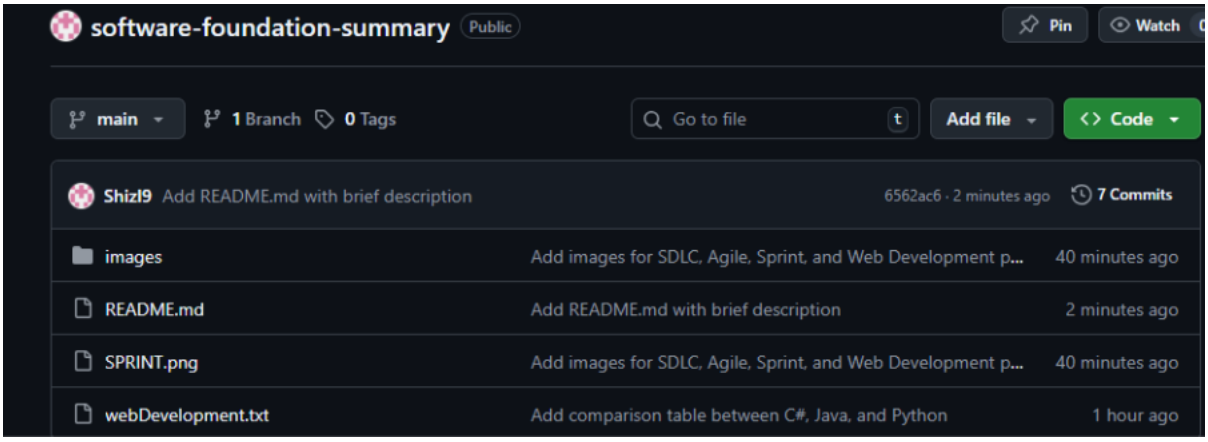
```
C:\Users\sheik\Downloads\software foundation summary>git add
 webDevelopment.txt

C:\Users\sheik\Downloads\software foundation summary>git com
mit -m "Add comparison table between C#, Java, and Python"
[master e7a70ed] Add comparison table between C#, Java, and
Python
 1 file changed, 11 insertions(+), 1 deletion(-)

C:\Users\sheik\Downloads\software foundation summary>
```




```
C:\Users\sheik\Downloads\software foundation summary>git add
 images

C:\Users\sheik\Downloads\software foundation summary>git com
mit -m "Add images for SDLC, Agile, Sprint, and Web Developm
ent parts"
[master a195522] Add images for SDLC, Agile, Sprint, and Web
 Development parts
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 images/sprint.png

C:\Users\sheik\Downloads\software foundation summary>
```



commit ,(add a README.md file (search how to add this file ,On the GitHub repo -10
the change on GitHub



write into into breif ,open the readme file ,pull the change from your local computer -11
,(description of all the content in the folder (describe what the folder's files cover briefly
commit this change and push

```
C:\Users\sheik\Downloads\software foundation summary>git pull
Already up to date.

C:\Users\sheik\Downloads\software foundation summary>
```

```
Software Foundation Summary#

:This folder contains

webDevelopment.txt** - Detailed explanation of frontend, backend, and database responsibilities, .NET **.١
.stack tools (SQL Server, C#, EF Core, LINQ, Blazor), and a comparison table between C#, Java, and Python
.images/** - Folder containing images illustrating SDLC, Agile, Sprint, and Web Development parts**.٢
.README.md** - This file giving a brief overview of the folder contents**.٣
```

```
C:\Users\sheik\Downloads\software foundation summary>git add
 README.md

C:\Users\sheik\Downloads\software foundation summary>git com
mit -m "Update README.md with brief description of folder co
ntents"
[main c2424a4] Update README.md with brief description of fo
lder contents
 1 file changed, 14 insertions(+), 4 deletions(-)

C:\Users\sheik\Downloads\software foundation summary>git pus
h
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 703 bytes | 140.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0
)
To https://github.com/Shizl9/software-foundation-summary.git
   6562ac6..c2424a4  main -> main

C:\Users\sheik\Downloads\software foundation summary>
```